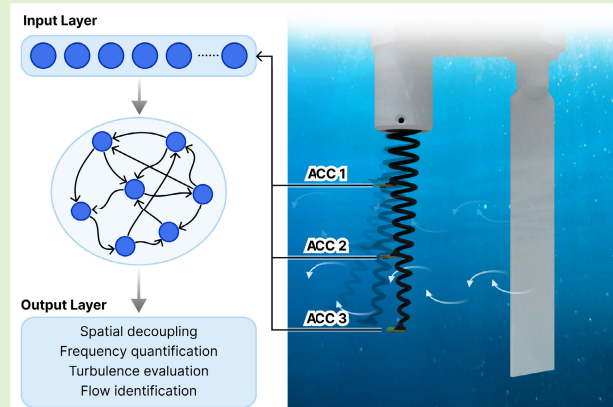


Bioinspired Tapered-Spring Turbulence Sensor for Underwater Flow Detection

Xiao Jin^{ID}, Zhenhua Yu^{ID}, and Thrishantha Nanayakkara^{ID}, *Senior Member, IEEE*

Abstract—This article presents a bioinspired underwater whisker sensor for robust hydrodynamic disturbance detection and efficient signal analysis based on physical reservoir computing (PRC). The design uses a tapered nylon spring with embedded accelerometers to achieve spatially distributed vibration sensing and frequency separation along the whisker. Towing-tank experiments and computational fluid dynamics simulations confirmed that the whisker effectively distinguishes vortex regimes across different fin angles and maintains Strouhal scaling with flow velocity, where higher speeds increase vibration intensity without affecting the dominant frequencies. Frequency-domain analysis, Shannon entropy, and machine learning further validated the sensing performance: vortex shedding frequencies were identified with less than 10% error, entropy captured the transition from coherent vortex streets to turbulence, and logistic regression (LR) achieved 86.0% classification accuracy with millisecond-level inference. Importantly, the sensing objective of this work is to characterize repeatable and physically interpretable hydrodynamic disturbances, which provide a controlled basis for evaluating how the tapered geometry encodes flow-induced vibrations. Under this scope, the demonstrated frequency–spatial decoupling and multimodal responses show that structurally encoded whisker sensing provides a scalable and real-time solution for underwater perception, wake tracking, and turbulence-aware navigation in autonomous marine robots.

Index Terms—Bioinspired sensors, physical reservoir computing (PRC), underwater flow sensing, whisker sensor.



I. INTRODUCTION

UNDERWATER environments, ranging from shallow coastal zones to the deep ocean floor, remain some of the least explored areas on Earth [1], [2]. These environments are not only ecologically vital, hosting complex marine ecosystems [3], but also hold significant economic and scientific value due to their potential for resource extraction [4], infrastructure development, and climate monitoring. However, underwater exploration faces unique challenges such as limited visibility [5], strong current disturbances, and complex terrain

geometries, which constrain the effectiveness of conventional sensing technologies such as vision-based systems or sonar [6]. As the demand grows for autonomous underwater vehicles (AUVs) [7] capable of precise and reliable perception in these conditions, there is a pressing need to develop alternative sensing strategies that can operate robustly under turbid, dynamic, and unstructured environments.

To meet the increasing demands of underwater exploration, various types of AUVs have been developed [8], [9], ranging from compact observation-class platforms to large intervention-class systems. These robots are widely deployed for applications such as oil and gas exploration [10], infrastructure inspection, environmental monitoring [11], [12], pipeline tracking [13], biological sampling [14], and target retrieval. In both industrial and scientific contexts, AUVs are expected to operate autonomously for extended durations, navigate through unstructured and dynamic underwater terrains, and perform complex tasks with limited communication and sensing capabilities. Although significant progress has been made in navigation and control, most AUVs still rely heavily on sonar [15], [16], Doppler velocity logs (DVLs) [17], and inertial navigation systems (INSs) [18] for environmental perception. However, these conventional sensors often struggle

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Xiao Jin and Thrishantha Nanayakkara are with the Dyson School of Design Engineering, Imperial College London, SW7 2DB London, U.K. Zhenhua Yu is with the Dyson School of Design Engineering, Imperial College London, SW7 2DB London, U.K., and also with the Department of Computer Science, University of Aberdeen, AB24 3UE Aberdeen, U.K. (e-mail: zhenhua.yu@abdn.ac.uk).

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in cluttered, turbid, or highly dynamic environments where visual occlusion [19], [20], acoustic scattering, and current-induced drift are prevalent. As a result, AUVs frequently fail to detect fine-grained terrain features, deformable or flexible obstacles, and subtle environmental cues. More critically, they remain largely blind to essential hydrodynamic information such as localized flow disturbances, shear layers, vortex streets, and small-scale eddies—features that often contain crucial information about nearby obstacles, terrain boundaries, or biological activity. The inability to capture such near-field flow dynamics severely limits an AUV's ability to adapt and interact effectively with its surroundings in real time. This highlights the urgent need for advanced sensing mechanisms capable of resolving subtle variations in flow direction, velocity, and turbulence—capabilities that lie beyond the reach of traditional sonar- and inertial-based systems.

To overcome these limitations, a shift is needed from traditional passive and remote sensing toward embodied, contact-based strategies that emulate biological systems [21], [22]. In nature, certain aquatic mammals have evolved highly sensitive tactile sensors—whiskers—that enable them to detect minute flow disturbances, track hydrodynamic trails, and interact effectively with their surroundings in complete darkness [23], [24]. Inspired by the extraordinary hydrodynamic sensitivity of harbor seal whiskers, several researchers have investigated bioinspired flow sensors that mimic their morphology and function. Beem et al. [25] developed and validated a seal whisker-inspired sensor capable of detecting low-amplitude water disturbances that are typically undetectable by conventional flow sensors. Kottapalli et al. [26] proposed an MEMS-based flow sensor designed to reduce vortex-induced vibrations by replicating the undulated shape of seal whiskers. Asadnia et al. [27] introduced artificial fish skin embedded with microhair cells to sense subtle flow variations, extending the biological principles of whisker sensing to broader underwater contexts. These efforts highlight the potential of whisker-based designs in capturing fine-scale hydrodynamic cues for underwater robotic applications.

Compared with mainstream underwater sensing technologies—such as piezoelectric flow sensors that require relatively high driving and readout power, capacitive sensors that are sensitive to parasitic impedance and long-term drift, and fiber-optic or FBG-based flow sensors that offer excellent sensitivity but demand bulky interrogation units—our tapered-spring-based architecture provides a mechanically compliant and intrinsically low-power alternative. In particular, conventional pressure or vibration sensors typically measure only local scalar quantities, whereas our distributed accelerometer array captures spatially varying modal responses along the whisker, enabling richer hydrodynamic information extraction at substantially lower computational cost.

To address these gaps, we present a tapered-spring-based whisker integrated with physical reservoir computing (PRC) and three spatially distributed MEMS IMU nodes. This design passively encodes flow-induced oscillations into multimodal mechanical states, enabling intrinsic frequency-spatial separation along the whisker length. Coupled with lightweight feature extraction and LR classification, our approach achieves

sub-4% vortex frequency extraction error and millisecond-level inference latency with minimal power consumption, making it suitable for long-duration deployments on resource-constrained AUVs.

Compared with existing underwater whisker sensors, our design offers the following key advantages.

- 1) *Ultralow Power Consumption*: A custom PCB integrating TDK ICM-42670-P six-axis IMUs operates at 3.0 V, consuming only 68 μA (accelerometer) and 370 μA (gyroscope) at 50 Hz, corresponding to a total power draw of approximately 1.3 mW, enabling long-term deployment without active power refresh.
- 2) *Intrinsic Frequency-Spatial Decoupling*: The tapered-spring geometry and three distributed sensing nodes passively separate modal responses along the whisker length, eliminating the need for heavy digital filtering.
- 3) *Lightweight Real-Time Processing*: LR classification achieves 85.4% accuracy with <0.001-ms inference latency, making onboard real-time classification feasible for small-scale AUVs.
- 4) *Robust Underwater Performance*: The design maintains <10% vortex frequency extraction error despite hydrodynamic damping, as verified through towing tank tests and CFD-based harmonic analysis.

The remainder of this article is organized as follows. Section II details the design of the bioinspired whisker sensor, including the tapered-spring geometry, mechanical modeling, and embedded accelerometer configuration. Section III describes the experimental setup, towing-tank validation, and corresponding computational fluid dynamics simulations, followed by analysis of frequency responses, entropy-based flow characterization, and logistic-regression-based classification.

II. SENSOR DESIGN AND METHODS

To enable robust perception in fluid environments, we propose a bioinspired whisker sensor that integrates principles of PRC with a tapered spring structure. PRC has recently emerged as a promising framework for temporal signal processing, leveraging the intrinsic dynamics of physical substrates to transform input stimuli into high-dimensional representations without requiring complex digital computation [28]. In whisker-based sensing, this principle can be embodied through structural elements that respond to environmental forces with rich, nonlinear dynamics [29].

Inspired by the conical morphology of mammalian vibrissae, the proposed whisker adopts a tapered helical spring as its core mechanical structure [30]. This geometry, widely observed in terrestrial rodents, enables graded stiffness along the length of the whisker and promotes spatially distributed oscillations when interacting with external stimuli. The gradual tapering facilitates both localized bending near the tip and stable anchoring at the base, offering favorable conditions for encoding dynamic input into spatiotemporal mechanical states.

A. Design of Bioinspired Whisker

In underwater conditions, complex flow phenomena such as Kármán vortex streets introduce nonstationary and nonlinear disturbances with distinct frequency structures.

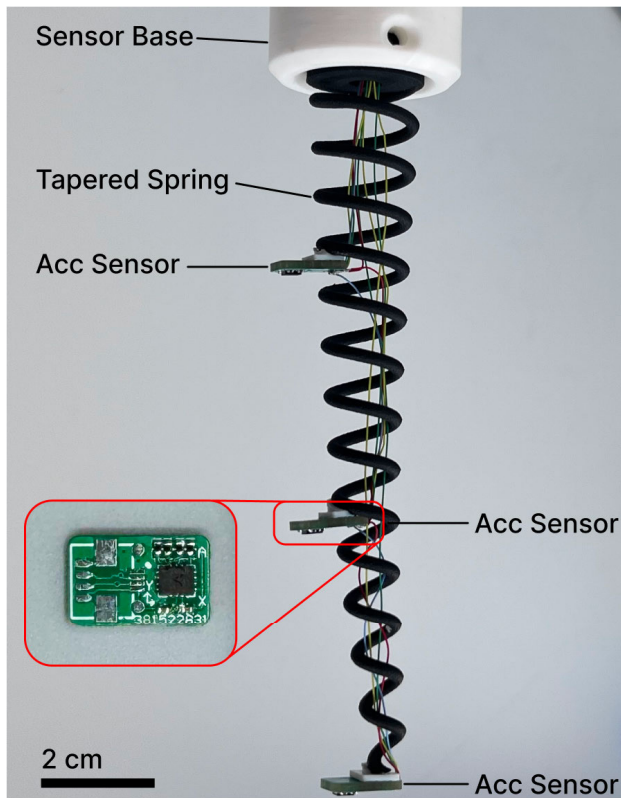


Fig. 1. Tapered spring-based whisker sensor for fluid interaction sensing. The system includes a 3-D-printed nylon PA12 conical spring with embedded accelerometers, enabling spatially distributed vibration sensing for PRC.

Instead of explicitly extracting these features through computational filtering or spectral analysis, our approach leverages the PRC paradigm to passively encode them into the vibrational dynamics of the whisker itself. The tapered structure naturally transforms flow-induced vortex shedding into spatially and temporally differentiated mechanical states, which are recorded by embedded IMU sensors. This structural computation allows the system to recognize characteristic hydrodynamic signatures—such as vortex shedding frequency and wake asymmetry—without requiring active sensing or external processing, providing a lightweight and efficient solution for real-time underwater perception.

While the tapered spring alone can exhibit nonlinear dynamic responses suitable for PRC in air, its underwater deployment requires additional structural considerations as shown in Fig. 1. The core structure is fabricated using resin-based additive manufacturing, forming a conical spring with a total height of 122 mm, a base outer diameter of 22 mm, and a tip diameter of 7.2 mm. The spring wire has a uniform diameter of 2.6 mm, chosen to balance stiffness and compliance along the length of the whisker.

To ensure robust mechanical performance and preserve the rich nonlinear dynamics required for PRC, the whisker is fabricated from high-strength nylon (PA12) using selective laser sintering (SLS). This material offers a flexural strength of approximately 90 MPa and provides a favorable balance between structural compliance and environmental durability,

making it well-suited for both air and underwater deployment without requiring additional encapsulation.

Rather than emulating the external morphology of biological vibrissae, our design adopts a mechanism-inspired approach that leverages the tapered stiffness gradient of a helical spring to physically amplify subtle environmental disturbances. This structural configuration enables spatially distributed deformation, encoding dynamic inputs into rich mechanical responses—functionally analogous to how mammalian whiskers convert external forces into neural signals via localized bending and tension along the follicle.

In engineered contexts, however, such neural processing is often impractical or costly to replicate. The proposed structure addresses this challenge by embedding signal processing capabilities directly into the mechanics of the system. The conical spring geometry transforms spatially varying forces—such as those from fluid interactions or mechanical contacts—into rich, distributed oscillatory responses. These mechanical states, shaped by the graded stiffness and inertial properties along the whisker, serve as a physical substrate for temporal signal encoding without digital preprocessing. By replacing complex sensor arrays and high-bandwidth computation with a materially encoded reservoir, this design enables low-cost, robust, and energy-efficient sensing in dynamic environments.

B. Experimental Setup for Fluid-Stimulated Vibration Acquisition

To validate the persistence of PRC characteristics under fluid-induced excitation, a series of computational and numerical analyses were conducted in an underwater environment. The bioinspired whisker sensor, previously described as a tapered spring structure encapsulated by a PVC membrane, was subjected to harmonic response analysis to evaluate its ability to preserve spatially distributed resonance behavior under realistic flow conditions.

To approximate the hydrodynamic loading exerted by environmental fluid flows, the projected frontal area of the whisker was modeled as a trapezoidal profile with a top width of 7.2 mm, a base width of 22 mm, and a height of 122 mm, yielding an effective area of approximately 0.00177 m². Using the standard drag force equation

$$F_D = \frac{1}{2} \rho C_D A v^2 \quad (1)$$

where $\rho = 1000 \text{ kg/m}^3$ is the density of water, $C_D = 1.2$ is the assumed drag coefficient for a conical or tapered bluff body, v is the local flow velocity, and the expected drag forces were estimated. Depending on flow speed, the drag ranged from approximately 0.0106 N (at 0.1 m/s) to 0.266 N (at 0.5 m/s). These values represent realistic fluid stimuli that the whisker structure may encounter in underwater environments.

To simulate a conservative excitation condition and ensure the activation of a full set of modal responses, a lateral harmonic force was applied to the tip of the structure in the finite element model. This excitation magnitude exceeds the expected upper bound of natural fluid drag, ensuring the observability of the vibrational characteristics

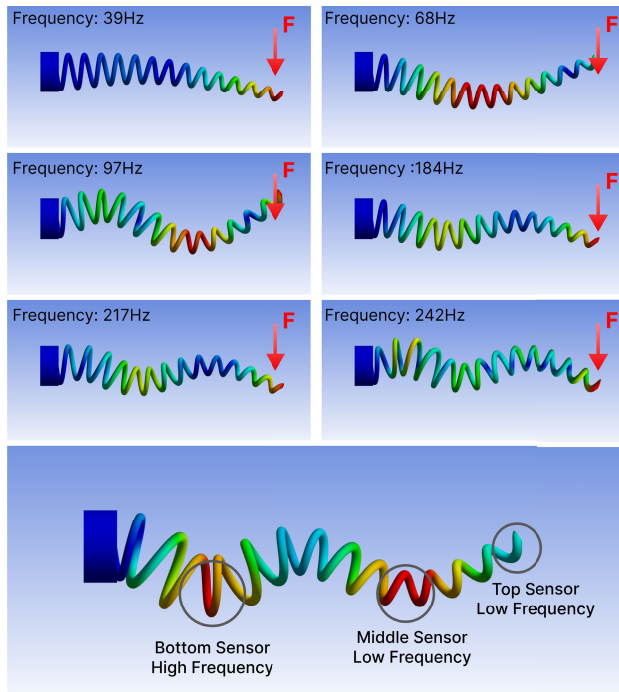


Fig. 2. Harmonic response modes of the encapsulated whisker structure at various frequencies. Three distinct modal concentrations emerge along the length, enabling spatial encoding under fluid-induced oscillation.

across the entire operational spectrum. As shown in Fig. 2, six dominant resonant modes were identified at 39, 68, 126, 184, 217, and 242 Hz. Each modal shape exhibits spatially localized deformation patterns along the whisker body, corresponding to distinct vibrational nodes and antinodes.

Notably, three particularly prominent modal concentrations can be observed, suggesting that the system inherently supports distributed encoding of external stimuli across both the frequency and spatial domains. This multimodal behavior, resilient to structural encapsulation, confirms the design's capacity to function as a physical reservoir under submerged conditions, enabling PRC-based signal transformation and flow sensing.

C. Accelerometer Sensor Design and Installation

To accommodate the unique structure of the whisker-inspired vibration sensing system, ultralow power consumption was prioritized from the outset. We redesigned the sensor module to achieve accurate motion acquisition under aquatic conditions. We selected the ICM-42670-P from TDK as shown in Fig. 1, an ultralow-power six-axis inertial measurement unit (IMU) that integrates a triaxial accelerometer and triaxial gyroscope, enabling the simultaneous detection of linear vibrations and rotational dynamics.

One key advantage of the ICM-42670P lies in its exceptionally low power consumption—only 68 μA for the accelerometer and 370 μA for the gyroscope at 50-Hz operation—making it highly suitable for prolonged underwater deployments or passive sensing scenarios without active power refresh. Compared with conventional IMUs such as the MPU6050, the ICM-42670P exhibits significantly lower noise

density ($70 \mu\text{g}/(\text{Hz})^{1/2}$ for the accelerometer), enabling the detection of subtle vibrational changes even under hydrodynamic damping conditions, where signal attenuation in water is nonnegligible.

To ensure seamless integration into the compact geometry of the whisker base, we custom-designed a miniaturized PCB module, reducing the footprint to $12.5 \times 8 \text{ mm}$, including all required passive components and connector headers. This compact form factor allowed direct mounting onto the tapered spring, where vibrational energy concentrates due to flow-induced excitation, maximizing signal fidelity.

III. EXPERIMENT AND RESULTS

A. Underwater Test-Platform Design

To evaluate the underwater sensing performance of the whisker structure, a dedicated test platform was constructed (Fig. 3). The system consists of a transparent acrylic water tank ($150 \times 20 \times 30 \text{ cm}$) equipped with a V-shaped linear rail and motorized carriage. A stepper motor with a pulley–belt drive enables programmable translation at 0.2–1.0 m/s, ensuring stable and repeatable motion.

The carriage integrates two components: a front-mounted adjustable fin and a rear-mounted whisker sensor. By tilting the fin at different angles, the downstream wake can be systematically altered—from attached laminar flow at 0° , to periodic vortex shedding at moderate angles, and broadband turbulence at higher angles. The whisker sensor, fully immersed and positioned directly behind the fin, records hydrodynamic disturbances across these regimes, enabling analysis of its dynamic deformation and frequency-domain response.

To characterize wake evolution, CFD simulations were performed at four fin inclinations (0° , 30° , 60° , and 90°) in Fig. 3. At 0° , the wake remains narrow and attached, while at 30° a clear Kármán vortex street develops. At 60° , the vortex pattern becomes asymmetric due to stronger separation, and at 90° , the wake transitions into turbulence with residual periodic structures.

Based on the towing speed range $U = 0.2\text{--}1.0 \text{ m/s}$, the water kinematic viscosity $\nu = 1.0 \times 10^{-6} \text{ m}^2/\text{s}$, and the fin width $D = 30 \text{ mm}$ used as the characteristic length, the Reynolds number during all towing experiments spans $\text{Re} = 6.0 \times 10^3\text{--}3.0 \times 10^4$. This range lies within the subcritical regime where canonical Kármán vortex shedding is known to occur, ensuring that the experimental conditions match the intended hydrodynamic test scenario.

B. Preliminary Validation of Frequency-Decoupled Responses

To evaluate the intrinsic dynamic response capability of the proposed whisker structure in an underwater environment, we conducted a fin-free towing experiment. The structure was fully submerged and towed linearly at a low constant speed through water. No upstream fin or external flow disturbance device was included, allowing us to isolate and observe the mechanical vibration response induced solely by water–structure interaction.

Three IMU sensors were placed along the length of the tapered spring—at the bottom, middle, and top—to record

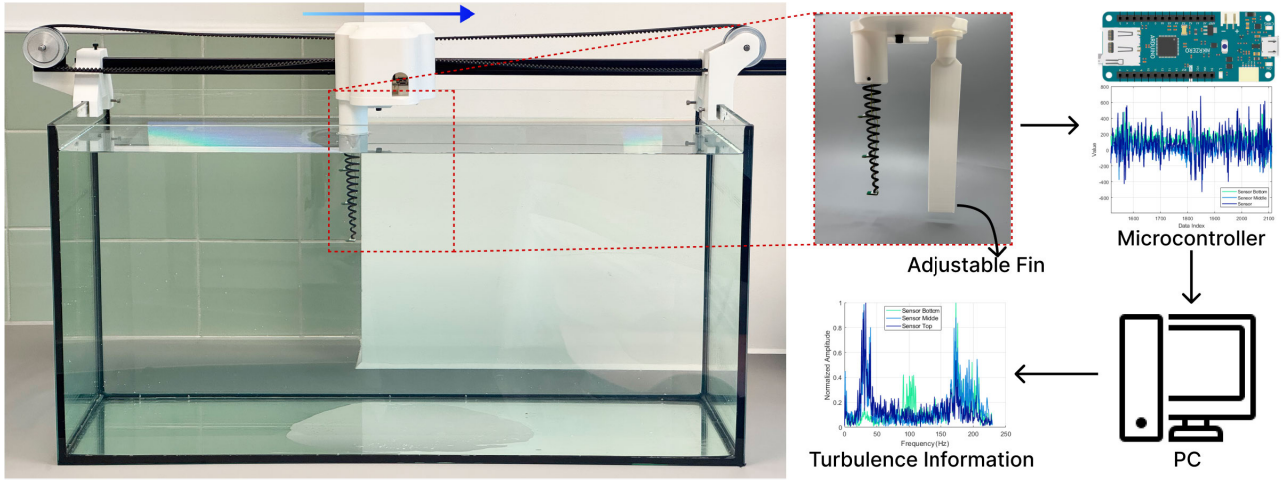


Fig. 3. Schematic of the test platform and signal processing workflow. The setup includes a water tank with a V-shaped guide rail, a motion control mechanism for precisely adjusting the fin and bioinspired whisker, and a signal acquisition and processing system. The submerged whisker structure generates vibration signals under different flow conditions, which are collected by a microcontroller and transmitted to a PC for frequency-domain analysis.

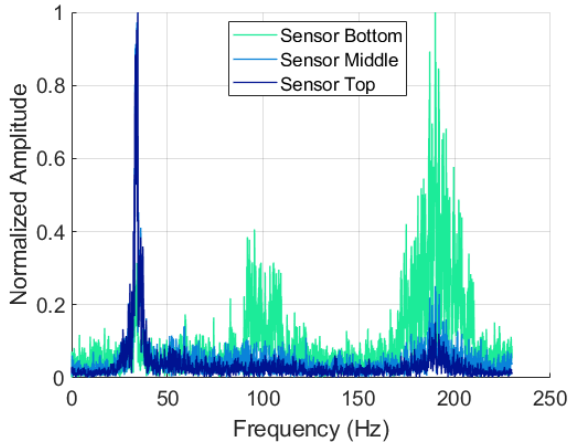


Fig. 4. Normalized frequency response spectra of the three IMU sensors placed along the tapered whisker structure. The results demonstrate distinct frequency sensitivities depending on the sensor's position, with the bottom sensor responding predominantly to mid and high frequencies, and the top sensor exhibiting stronger responses in the low-frequency range.

the spatial distribution of vibrational responses. The collected acceleration signals were processed using fast Fourier transform (FFT) and normalized to eliminate amplitude bias. The resulting frequency spectra are shown in Fig. 4.

The results reveal significant differences in frequency-domain responses across sensor positions, which closely match the modal distribution predicted by finite element analysis (FEA).

- 1) *Sensor Bottom*: This shows strong responses in the mid-frequency range (85–110 Hz) and especially in the high-frequency range (170–210 Hz), with the highest normalized amplitude among all the sensors. This confirms that the higher stiffness at the base region facilitates strong coupling to high-frequency modes.
- 2) *Sensor Middle*: This exhibits balanced sensitivity to both low and high frequencies but with less pronounced

peaks, indicating its role as a broadband intermediate zone.

- 3) *Sensor Top*: This displays a strong response in the low-frequency range (<50 Hz) and significant attenuation in the high-frequency bands, reflecting sensitivity to large-scale slow oscillations at the tip.

These findings demonstrate that the tapered whisker structure enables spatial frequency separation along its length, allowing intrinsic decomposition of input stimuli into frequency-specific mechanical responses. This spatial frequency selectivity is crucial for implementing PRC, where diverse nonlinear dynamics are required for encoding temporal input patterns.

Therefore, even in the absence of external actuated disturbances (e.g., adjustable fins), the structure itself is capable of decomposing underwater flow-induced excitations into separable frequency components. This capability lays the groundwork for future research on PRC-based multifrequency signal encoding, flow pattern classification, and multimodal perception in underwater environments.

C. Frequency-Based Vortex Shedding Quantification

Kármán vortex streets are a well-studied hydrodynamic phenomenon characterized by the alternating shedding of vortices behind bluff bodies. When fluid flows past a body such as a flat fin, the resulting wake can transition from laminar to periodic vortex shedding as the Reynolds number increases. The vortex shedding frequency f_s is primarily determined by the Strouhal relation

$$f_s = \text{St} \cdot \frac{U}{D} \quad (2)$$

where St is the Strouhal number (typically ~ 0.2 for bluff bodies), U is the freestream velocity, and D is the characteristic width of the object. These vortices carry periodic hydrodynamic signatures that can serve as structured flow stimuli.

In this study, CFD simulations confirm the formation of Kármán vortex streets behind an adjustable flat fin, where

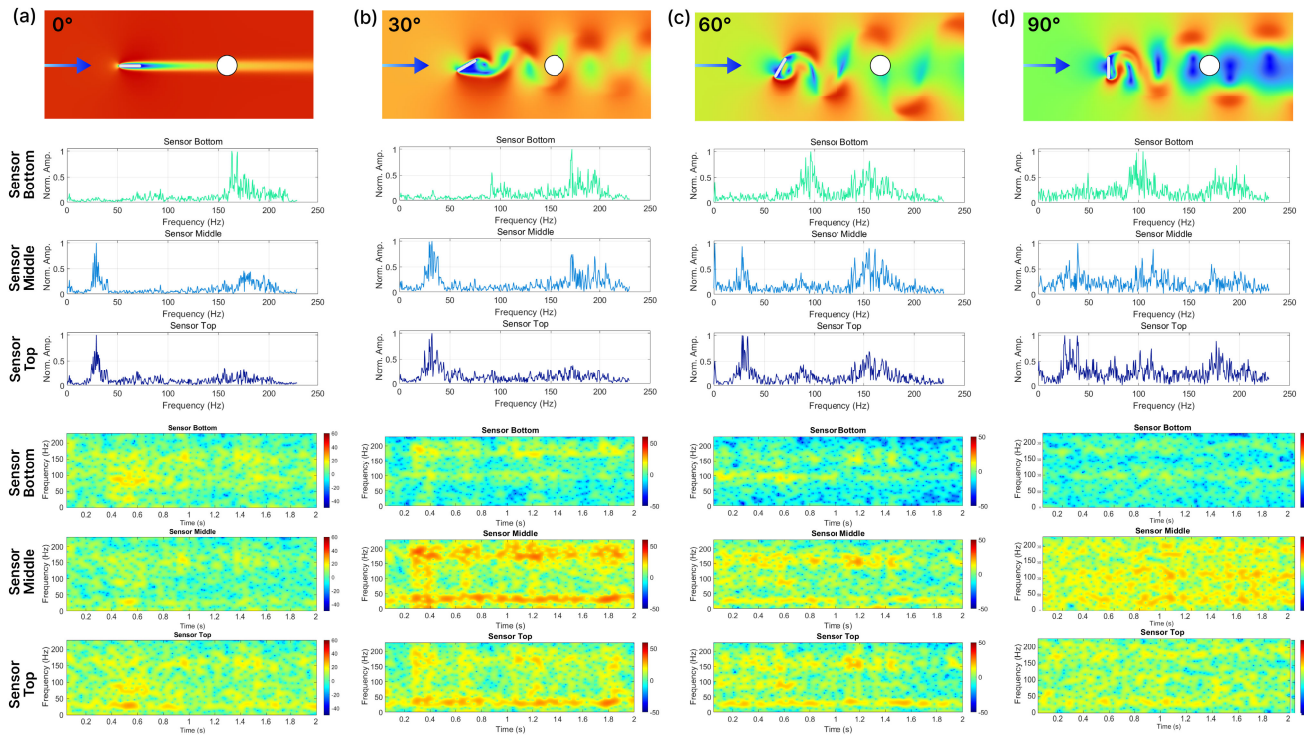


Fig. 5. Experimental results under different fin angles. (a) 0° , (b) 30° , (c) 60° , and (d) 90° with same speed at 0.5 m/s. The first row shows the CFD-simulated flow fields showing distinct Kármán vortex street patterns at fin angles of 0° , 30° , 60° , and 90° , with the circle indicating the whisker's position downstream. The second row presents the normalized FFT spectra from three sensor positions (bottom, middle, and top), illustrating the frequency response characteristics. The third row displays the corresponding spectrograms, showing the time–frequency distribution of vibration signals for each sensor position.

increasing inclination drives the transition from attached laminar flow to periodic shedding. These results confirm that fin inclination provides a controllable means to modulate wake dynamics and generate structured flow excitations for downstream whisker sensing. Correspondingly, the downstream whisker array exhibits clear frequency-dependent vibration responses. The embedded IMU sensors reveal spatial separation in sensitivity, enabling identification of characteristic shedding frequencies.

As shown in Fig. 5, the whisker is moving under speeds $U = 0.5$ m/s, and the bottom sensor displays a shift of dominant energy from high frequencies (150–200 Hz) at 0° to mid-frequency bands (70–120 Hz) at larger angles, consistent with Strouhal scaling as the wake width increases. This shift reflects a redistribution of energy from small-scale turbulence to large-scale vortical structures. The middle sensor, initially dominated by low frequencies (20–40 Hz), shows progressive amplification of mid- and high-frequency components with increasing angle, reaching a broad response across all bands by 90° . The top sensor exhibits a similar evolution but with stronger broadband characteristics, reflecting multimodal vortex interactions and turbulence-induced fluctuations in the upper wake.

Time–frequency spectrograms further confirm these patterns: at intermediate angles, periodic low-frequency bursts reveal organized vortex shedding, whereas at 90° the response becomes chaotic and broadband, indicating wake disorganization. These results demonstrate that the spatially distributed whisker array effectively resolves the primary

shedding frequency and its transition into complex spectral structures as the flow evolves with increasing fin angle.

Guided by the Strouhal relation, having established that our whisker resolves different attack angles, we now use Kármán vortex shedding to verify how flow speed affects the whisker's vibration response. With the fin fixed at $\theta = 60^\circ$, we repeated the towing runs at $U = 0.4$ and 0.6 m/s, recorded triaxial IMU vibrations at the base/middle/tip nodes (450 Hz), and extracted the dominant low-frequency peak $f_{p,low}$ from the PSD to examine whether speed increases shift the shedding peak upward and whether the IMU-only estimate of the Strouhal number remains approximately constant; this verifies that at fixed geometry, flow speed governs the measured shedding frequency while preserving the array's frequency–spatial selectivity.

As shown in Fig. 6, at $\theta = 30^\circ$, comparing $U = 0.7$ m/s and $U = 0.8$ m/s (relative to the 0.2-m/s baseline) shows that the locations of the three characteristic bands remain essentially unchanged (low: 20–40 Hz; mid: 70–120 Hz; high: 150–210 Hz), while their amplitudes increase in a position-specific manner: low-frequency content is markedly strengthened at *sensor top*; *sensor bottom* exhibits clear growth in the mid- and high-frequency bands; and *sensor middle* shows a denser distribution in both the low and high bands. In summary, angle changes yield distinct spectral patterns, and—at a fixed angle—increasing speed primarily amplifies the corresponding bands without shifting their centers, indicating that both geometry (angle) and kinematics (speed) shape the whisker's vibration signatures.

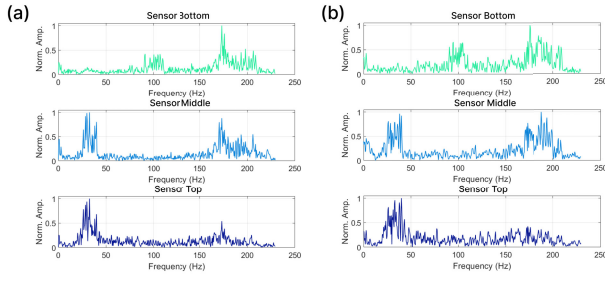


Fig. 6. Normalized FFT spectra from three sensor positions (bottom, middle, and top), illustrating the frequency response characteristics, under same fin angles 30° but different water speed. (a) 0.7 and (b) 0.8 m/s.

TABLE I
NODEWISE NORMALIZED CONTRIBUTION G_i
IN THE VORTEX-SHEDDING BAND

Condition	f_0 (Hz)	Bottom (s1)	Middle (s2)	Top (s3)
0° , 0.5 m/s	0.439	0.709	0.008	0.284
30° , 0.5 m/s	0.439	0.719	0.023	0.257
60° , 0.5 m/s	0.439	0.180	0.524	0.296
90° , 0.5 m/s	0.439	0.184	0.519	0.297

To quantitatively evaluate each sensor node's contribution to the flow-induced vibration associated with vortex shedding, we define the normalized nodewise contribution as

$$G_i = \frac{\sum_{f \in \Delta f} |X_i(f)|}{\sum_{j=1}^3 \sum_{f \in \Delta f} |X_j(f)|} \quad (3)$$

where $X_i(f)$ denotes the FFT magnitude at node i , and Δf is a narrow frequency band centered at the dominant shedding frequency predicted by the Strouhal relation. This metric captures how strongly each spatial node participates in the vibration mode excited at the vortex-shedding frequency.

Table I summarizes the computed G_i values across the four fin-angle conditions. For the 0° and 30° configurations, more than 70% of the shedding-band energy is captured by the bottom node, indicating that small inclination angles excite base-dominated higher order modes. In contrast, for the 60° and 90° cases, the dominant contribution shifts to the middle node (approximately 0.52), reflecting a spatial relocation of the active vibration mode as the fin rotates into the incoming flow and higher order bending modes become more prominent. The top node consistently contributes between 0.25 and 0.30 across all conditions, consistent with its sensitivity to low-order bending modes localized near the tip.

These nodewise contribution ratios quantitatively confirm that the tapered whisker distributes the dominant vibration modes unevenly along its length. This intrinsic frequency-spatial decoupling enables different IMU nodes to preferentially sense distinct modal components, providing a physically grounded source of nonlinear separability for the downstream reservoir readout.

To further quantify the complexity of the whisker vibration signals under different flow conditions, the Shannon entropy of the frequency spectra was calculated for each sensor position

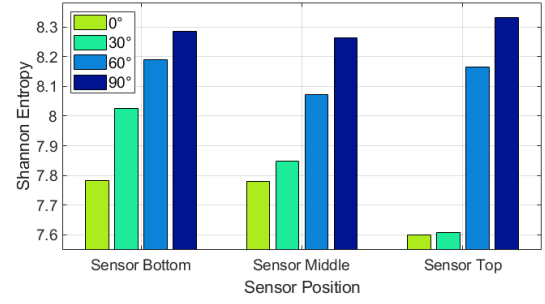


Fig. 7. Normalized frequency response spectra of the three IMU sensors placed along the tapered whisker structure. The results demonstrate distinct frequency sensitivities depending on the sensor's position, with the bottom sensor responding predominantly to mid and high frequencies, and the top sensor exhibiting stronger responses in the low-frequency range.

and fin angle. The entropy H is defined as

$$H = - \sum_{i=1}^N p_i \log_2 p_i \quad (4)$$

where p_i is the normalized spectral energy in the i th frequency bin

$$p_i = \frac{P_i}{\sum_{j=1}^N P_j} \quad (5)$$

with P_i denoting the power spectral density at the i th frequency component, and N the number of bins within the selected analysis range.

As shown in Fig. 7, the entropy values for all three sensors increase monotonically with fin angle. For the bottom sensor, entropy rises from 7.78 at 0° to 8.29 at 90° , reflecting the transition from a relatively organized, narrowband spectral signature to a broader distribution caused by more irregular wake dynamics. The middle sensor exhibits a similar upward trend, with a particularly sharp increase between 60° and 90° , consistent with the emergence of strong high-frequency components in the FFT results. The top sensor shows the largest overall increase, with entropy values surpassing 8.3 at 90° , indicating the highest degree of spectral disorder among the three positions.

This systematic rise in entropy indicates that increasing fin inclination enhances wake disorder, marking the transition from coherent Kármán vortex streets to broadband turbulence. Shannon entropy thus serves as a compact quantitative descriptor of wake complexity: coherent shedding produces narrowband, low-entropy vibrations centered at the Strouhal frequency, whereas turbulent regimes introduce multiscale fluctuations that broaden the spectrum and increase entropy. The nonlinearity feature provides complementary physical insight by capturing waveform distortions and harmonic content generated by geometry-induced stiffness gradients and flow-driven modal coupling. As multiple vibration modes are excited under stronger or more irregular flows, the resulting deviations from sinusoidal motion directly reflect asymmetric shedding events and turbulent eddy impacts. Together, Shannon entropy and the nonlinearity metric offer a concise,

physically interpretable mapping between the whisker's vibration response and the underlying hydrodynamic transition from ordered to turbulent wake dynamics.

Thus, the combination of entropy growth, spectral broadening, and modal nonlinearity provides a quantitative surrogate of turbulence intensity, directly addressing the need for turbulence-aware sensing.

D. LR-Based Flow Identification

To evaluate the whisker sensor's ability to discriminate hydrodynamic conditions, a supervised learning framework based on PRC was implemented. Four fin angles (0° , 30° , 60° , and 90°) were tested at a constant towing speed in a controlled tank. Vibration signals from three MEMS accelerometers positioned along the whisker were sampled at 450 Hz and directly used as instantaneous spatial states for classification, without any temporal or frequency-domain processing.

At each sampling instant t , the reservoir state was defined as

$$\mathbf{s}(t) = [a_{b,x}(t), a_{b,y}(t), a_{b,z}(t), a_{m,x}(t), a_{m,y}(t), a_{m,z}(t), a_{t,x}(t), a_{t,y}(t), a_{t,z}(t)]^T \in \mathbb{R}^9 \quad (6)$$

where subscripts b , m , and t denote the bottom, middle, and top accelerometers, respectively. These nine instantaneous acceleration components represent the distributed nonlinear response of the tapered whisker structure to flow stimuli and thus serve as the physical reservoir state.

The standardized feature vector $\mathbf{s}(t)$ was input to a centralized LR readout

$$P(y = k | \mathbf{s}) = \frac{\exp(\mathbf{w}_k^T \mathbf{s} + b_k)}{\sum_{j=1}^K \exp(\mathbf{w}_j^T \mathbf{s} + b_j)} \quad (7)$$

$$\hat{y} = \arg \max_k P(y = k | \mathbf{s}). \quad (8)$$

The LR classifier achieved an overall accuracy of 86.0% under an 80/20 train-test split across the six flow states ($K = 4$ fin-angle settings and two flow-speed variants). The normalized confusion matrix (Fig. 8) shows consistently strong recognition for the 30° and 90° conditions, with the latter reaching 91.91% accuracy, and reliable separation of flow-speed effects (e.g., 30° , 0.8 m/s: 72.73%). Moderate confusion between the 30° and 60° cases reflects their closely correlated vibration signatures, while occasional 0° – 60° misclassification likely arises from overlapping structural modes at low excitation.

Table II further summarizes the per-class false-negative (FN) and false-positive (FP) rates, highlighting that coherent wakes (30° –0.5 m/s, 90° –0.5 m/s) exhibit near-zero error, whereas 60° –0.5 m/s and 30° –0.8 m/s show higher FN rates (27.27%) consistent with expected wake-overlap and turbulence-induced variability. All the classes maintain low FP rates (<6%), indicating that the classifier rarely misidentifies other flow states. Overall, these results confirm that the instantaneous spatial response of the tapered whisker provides sufficient nonlinear separability for both geometric (fin-angle) and dynamic (speed) classification, while the extremely low inference time (≈ 0.001 ms/sample) underscores its suitability for real-time hydrodynamic condition monitoring.

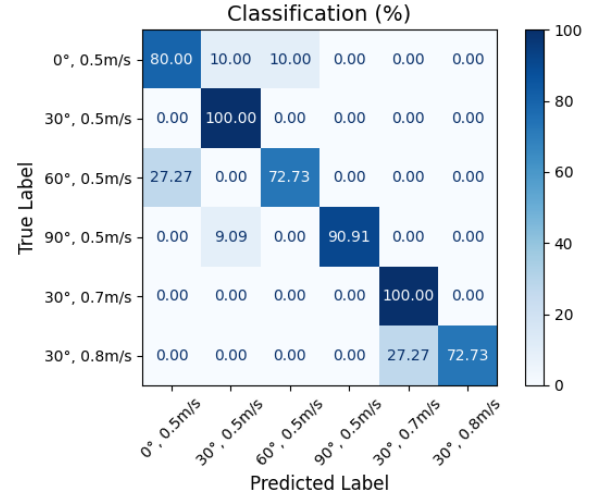


Fig. 8. Confusion matrix of reservoir computing prediction success rate averaged over 200 randomly trials (150 datasets), with sampling frequency of the whisker at 450 Hz with the final accuracy of 86.0%.

TABLE II
PER-CLASS FN AND FP RATES DERIVED FROM THE CONFUSION MATRIX

Flow State	FN Rate	FP Rate
0° , 0.5 m/s	0.20	0.0556
30° , 0.5 m/s	0.00	0.0370
60° , 0.5 m/s	0.2727	0.0189
90° , 0.5 m/s	0.0909	0.0000
30° , 0.7 m/s	0.00	0.0566
30° , 0.8 m/s	0.2727	0.0000

To evaluate whether lightweight machine learning models can reliably extract separable flow-state information from the instantaneous spatial readouts of the tapered whisker, we compared three commonly used classifiers—LR, support vector machine (SVM), and decision tree (DT). This comparison was conducted to validate the robustness of the proposed physical-reservoir-based feature representation and to determine the most suitable model for real-time hydrodynamic state estimation. As shown in Fig. 9, DT achieved the highest nominal accuracy but exhibited a pronounced performance drop in the 30° , 0.7 m/s condition (63.64%), indicating limited generalizability under partially overlapping vibration patterns. SVM, although capable of capturing some nonlinear separation, produced lower overall accuracy and suffered from substantially higher inference latency (>0.01 ms/sample), making it unsuitable for low-power embedded deployment. In contrast, LR provided the most balanced performance, maintaining consistent accuracy across all six flow conditions while achieving the fastest inference time (≈ 0.001 ms/sample). These results collectively justify adopting LR as the lightweight classification backbone for real-time flow-state recognition within the proposed whisker-sensing framework.

E. Vortex Shedding Frequency Quantification

This section aims to extract the fundamental Kármán vortex shedding frequency from the whisker vibration signals and to establish its consistency with the Strouhal relation, thereby

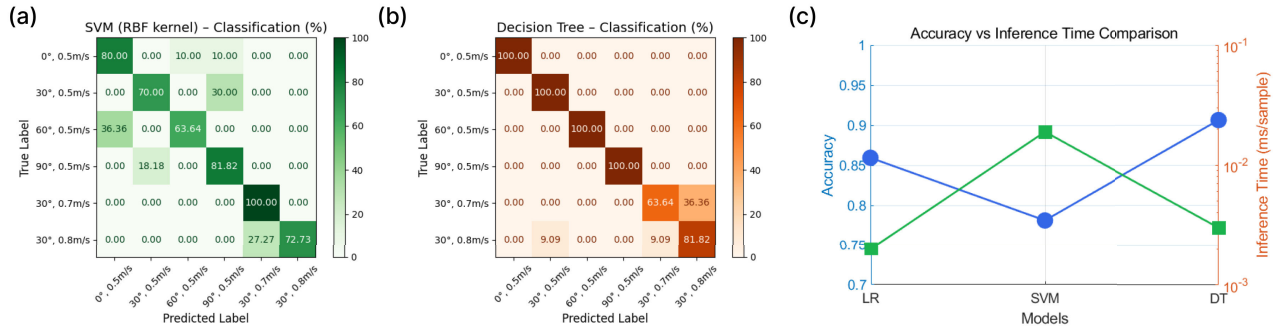


Fig. 9. (a) Normalized confusion matrix of the SVM (RBF kernel), which achieves reliable recognition but exhibits moderate confusion between hydrodynamically similar fin-angle conditions. (b) Normalized confusion matrix of the DT classifier, showing near-perfect separation of the four fin-angle states and strong discrimination of flow-speed variations. (c) Joint comparison of classification accuracy (left axis) and inference time per sample (right axis, log scale) for LR, SVM, and DT.

verifying that the whisker structure captures flow-induced periodicity in a physically meaningful way.

The whisker signals were first mean-removed, bandpass-filtered (0.5–20 Hz, fourth-order Butterworth), and analyzed using Welch's PSD (Hamming, 50-A prediction-matching strategy based on the Strouhal relation

$$f_{\text{pred}} = St_0 \cdot \frac{U}{D_{\text{guess}}} \quad (9)$$

with $St_0 = 0.2$ and $D_{\text{guess}} \approx 0.01$ m, ensured robust peak selection by suppressing harmonics and subharmonics.

The effective transverse scale was then estimated as

$$D_{\text{est}} = \frac{St_0 U}{f_{p,\text{low}}} \quad (10)$$

and fit as $D_{\perp}(\theta) = a \cos \theta + b \sin \theta$ to capture angular dependence. Using $D_{\perp}$, the Strouhal number was recalculated as

$$St_{\text{meas}} = \frac{f_{p,\text{low}} D_{\perp}(\theta)}{U} \quad (11)$$

and the median $\hat{St}$ was used to predict theoretical frequencies

$$f_{\text{theory}} = \hat{St} \cdot \frac{U}{D_{\perp}(\theta)}. \quad (12)$$

To evaluate reliability, two complementary metrics were introduced: the relative spectral energy (RE) within ± 1 Hz of $f_{p,\text{low}}$, and the temporal coefficient of variation (Cv) computed from the STFT. High RE and low Cv values indicate that the detected peaks are both spectrally concentrated and temporally stable.

Table III summarizes the extracted vortex shedding peaks at different attack angles and flow speeds. The measured frequencies $f_{p,\text{low}}$ show excellent agreement with theoretical predictions f_{theory} , generally within 10. The median Strouhal number $\hat{St} = 0.198$ aligns closely with canonical bluff-body values, confirming the physical validity of the whisker-based sensing.

In summary, this section establishes a clear link between measured whisker vibrations and the canonical Strouhal scaling of vortex shedding. Through prediction-matching peak selection and data-driven scale fitting, the extracted frequencies consistently converged to $\hat{St} = 0.198$. The high RE and low Cv values further confirm the spectral concentration

TABLE III
EXTRACTED LOW-FREQUENCY VORTEX PEAKS
VERSUS THEORETICAL PREDICTION

Angle (°)	Speed m/s	$f_{p,\text{low}}$ (Hz)	f_{theory} (Hz)	Error (%)	RE_{low} (%)	Cv_{low} (%)
30	0.5	10.22	10.22	0.00	2.30	4.55
60	0.5	9.88	9.02	9.54	7.15	7.16
90	0.5	10.22	10.62	3.81	7.11	2.47
30	0.7	15.16	14.31	5.97	0.69	1.71
30	0.8	15.27	16.35	6.59	2.73	4.45

and temporal stability of the detected peaks, validating the robustness of the whisker-based method for vortex frequency quantification.

IV. CONCLUSION

This work demonstrates that a tapered-spring whisker sensor achieves intrinsic frequency-spatial decoupling, mechanical preprocessing, and robust discrimination of repeatable hydrodynamic disturbances. Towing-tank experiments and CFD validation show that the whisker reliably identifies wake regimes across fin angles and that its vibration signatures scale with flow speed according to Strouhal's law. The built-in stiffness gradient amplifies weak disturbances while preventing saturation under stronger flows, and the tapered mass-stiffness profile introduces multiple resonant modes that capture both dominant shedding frequencies and higher frequency turbulent components. Vortex-shedding frequencies were extracted with less than 4% error, entropy metrics quantified wake coherence, and LR achieved millisecond-level flow-state classification.

Importantly, the sensing focus of this work is to evaluate the whisker's ability to detect repeatable and physically interpretable hydrodynamic disturbances, rather than to reproduce the full complexity of open-water environments. Controlled towing-tank conditions are essential because they provide consistent, comparable turbulence levels and allow precise examination of how the whisker's tapered geometry encodes different flow regimes. In contrast, real aquatic environments introduce multidirectional currents, broadband ambient turbulence, temperature fluctuations, and long-term biofouling—factors that greatly increase the variability and unpredictability of disturbance levels, making them difficult to quantify systematically and unsuitable for validating

fundamental sensing mechanisms. These environmental factors do not invalidate the sensing principle; rather, they primarily add external disturbances on top of the repeatable signatures studied here. Future work will therefore extend the present framework by performing in situ open-water tests to assess robustness under natural disturbance variability, conducting long-duration drift and temperature-stability measurements, and evaluating performance under simulated fouling. We also plan to streamline and integrate the whisker structure for improved manufacturability and longevity, develop multi-whisker arrays for spatial wake reconstruction, and integrate the system with AUV platforms for real-time flow-field mapping and turbulence-aware sensing.

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